

# Roots to Wings

Integrating Indian Knowledge Systems in Modern Teacher Education

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A policy and pedagogical analysis exploring how NEP 2020 bridges indigenous intellectual heritage with contemporary teacher preparation, holistic assessment, and inclusive classroom pedagogy.

# Rootedness balances cultural origins with modern global progress

Identity construction in a globalised educational landscape

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In a globalized world where students are constantly exposed to external technologies and philosophies, knowing one's cultural and intellectual heritage is imperative for grounded identity construction.

**Roots:** Indian philosophy, ethics, eco-friendly lifestyle, and social coexistence.

**Wings:** Modern science, mathematics, technology, and global knowledge frameworks.

**NEP 2020 serves as the intermediary, linking ancient Indian wisdom with the requirements of the modern classroom.**

# The Indian Knowledge System spans millennia of intellectual traditions

An interconnected web of wisdom, science, and pedagogy

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**Linguistics & Philosophy:** The Vedas, Upanishads, and Panini's grammar establishing formal structural linguistics.

**Mathematics & Astronomy:** Aryabhata's place value system, Brahmagupta's formalisation of zero, and algorithmic trigonometry.

**Health & Healing:** Charaka Samhita internal medicine and Sushruta Samhita anatomical surgical techniques.

**Institutional Centers:** Nalanda, Takshashila, and Vallabhi as premier interdisciplinary research universities fostering debate and inquiry.

# Modern philosophical titans championed holistic, child-centered learning

Foundations of modern Indian educational reform

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**Ishwar Chandra Vidyasagar:** Inclusive education, social reform, simplification of learning materials.

**Swami Vivekananda:** Character building, man-making education, manifestation of latent perfection.

**Rabindranath Tagore:** Joyful, stress-free learning in harmony with nature; freedom from rigid testing.

**Mahatma Gandhi:** Nai Talim (Basic Education), dignity of manual labor, productive self-reliance.

**Sri Aurobindo:** Integral Education developing physical, vital, mental, psychic, and spiritual dimensions.

**J. Krishnamurti:** Freedom from fear and comparison, deep self-awareness, dialogue without dogma.

# Decolonisation restores epistemic balance and plurality

Moving beyond symbolic gestures toward critical scholarly integration

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Colonial education introduced bureaucratic efficiency but marginalized indigenous ways of knowing. Decolonisation in teacher education does not mean emotional rejection of Western thought; it requires restoring epistemic balance where indigenous and global methodologies engage in mutually respectful dialogue.

**Indian Philosophical Foundations**  
Contextual ethics & holistic ontology



**Integrated Epistemic Balance**  
**Mutual Rigor & Coexistence**



**Western Pedagogical Sciences**  
Empirical research & methodologies

# Epistemic plurality is a recognized global educational movement

International alignment of culturally responsive pedagogy

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## Africa

Curriculum reforms incorporating Ubuntu and indigenous science to decenter exclusively Eurocentric epistemologies.

## Australia & Canada

Mandating First Nations and Aboriginal perspectives into teacher education standards to foster culturally safe classrooms.

## India (NEP 2020)

Integration of IKS across school and higher education for cultural rootedness, multilingualism, and planetary ethics.

# NEP 2020 and NCF 2023 structurally align education with developmental stages

5+3+3+4 Curricular architecture and holistic progress

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**Foundational (5 yrs):** Ages 3–8 · Play-based, ECCE, mother tongue

**Preparatory (3 yrs):** Ages 8–11 · Activity-based, FLN literacy & numeracy

**Middle (3 yrs):** Ages 11–14 · Experiential learning & vocational arts

**Secondary (4 yrs):** Ages 14–18 · Multidisciplinary choice & critical inquiry

## Panchakosha & Holistic Assessment

- **Annamaya:** Physical development & health
- **Pranamaya:** Vital energy & breath awareness
- **Manomaya:** Emotional and mental equilibrium
- **Vijnanamaya:** Intellectual discernment & inquiry
- **Anandamaya:** Aesthetic joy and spiritual peace
- **Holistic Progress Card (HPC):** 360-degree assessment by self, peer, and teacher.

# Educational reform relies entirely on the empowered teacher

Teachers as the transformative agents of change

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Policies and curriculum frameworks provide directional architecture, but authentic transformation occurs only through the educator. NEP 2020 firmly positions the teacher at the epicenter of educational renewal.

**IKS is not merely supplementary content; it is a fundamental way of thinking.**

Teacher education provides the rigorous academic structure, while Indian Knowledge Systems bring ethical grounding, mindfulness, and ecological consciousness.



# Integration requires embedding IKS organically across core disciplines

Moving away from isolated electives to unified curriculum design

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## Educational Psychology

Incorporating yogic mind theory, pranayama for emotional regulation, attention span research, and contemplative pedagogy.

## Environmental Studies

Embedding traditional water harvesting (bawlis, stepwells), sacred groves, tribal conservation ethics, and circular sustainability.

## Value Education & Polity

Connecting ancient democratic institutions (Sabha, Samiti, Gana) with constitutional duties and civic responsibility.

# Authentic integration relies on prepared faculty and community partnerships

Bridging institutional theory with living community traditions

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## Faculty Professional Development

Educators must engage with primary IKS texts through critical scholarship rather than dogma. Interdisciplinary workshops, translation labs, and collaborative seminars build academic grounding.

## Community & Artisan Collaboration

Inviting traditional craftspeople, local historians, ayurvedic practitioners, and grassroots ecologists into teacher preparation bridges the divide between textbook theory and living practice.

# Assessment and language must adapt to reflect multidimensional growth

Moving from high-stakes rote testing to holistic reflection

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## Reforming Exam Culture

Dismantling reliance on one-shot summative recall exams. Adopting reflective journals, portfolio assessment, peer dialogue, and field documentation.

## Multilingual Engagement

Indian languages carry distinctive pedagogical and epistemic frameworks. Encouraging mother tongue instruction and multilingual translation unlocks conceptual depth.

# Current implementation remains transitional and uneven

Empirical research findings on teacher education institutional reality

| Dimension         | Observed Institutional Gap                                                | Research Citation                          |
|-------------------|---------------------------------------------------------------------------|--------------------------------------------|
| Curriculum        | IKS relegated to isolated electives rather than core infusion             | <a href="#">Mahesh et al. (2023)</a>       |
| Faculty Training  | Scarce structured pedagogical orientation for teacher educators           | <a href="#">Dyer et al. (2004)</a>         |
| Curriculum Reform | Inertia and lack of validated multilingual instructional resources        | <a href="#">Booi &amp; Khuzwayo (2018)</a> |
| Community Ties    | Absence of formal recognition for non-formal indigenous knowledge holders | <a href="#">Tailby et al. (2020)</a>       |

# Structural and academic roadblocks threaten meaningful reform

Four key barriers to sustainable IKS embedding

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## Faculty Preparedness

Educators trained in monocultural Western frameworks lack exposure to IKS primary texts.



## Risk of Tokenism

Superficial mention in introductory chapters without systemic realignment of pedagogical methods.



## Academic Rigor

Avoiding uncritical celebration on one side and ideological dismissal on the other through strict scholarship.



## Resource Scarcity

Lack of accessible translations, peer-reviewed courseware, and open digital repositories.

# Systemic redesign and institutional incentives pave the way forward

Five interlocking institutional imperatives

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- 1 Embed, Do Not Append:** Integrate IKS perspectives organically into syllabus units of all foundational papers.
- 2 Institutional Grants & Support:** Fund research in local knowledge systems and curriculum innovation labs.
- 3 Develop Authentic OER:** Produce peer-reviewed open educational resources and bilingual glossaries.
- 4 Structured In-Service Orientation:** Conduct continuous professional development modules for teacher educators.

# A grounded education system nurtures the whole individual for a global future

Concluding vision for reflective, rooted teacher education

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Integrating Indian Knowledge Systems re-explores the enduring purpose of education in India: shaping character, ethical sensitivity, and social responsibility.

**"Rootedness ensures that our education system does not merely produce examination performers, but holistically developed citizens who engage confidently with global ideas from a foundation of deep intellectual heritage."**